

Progress Report: Sprint 12

CS.028 Optimizing Research Software Codes - 13 April 2026

[Kathryn Butler](#), [Joseph Schaab](#), [Michael McAllister](#)

Summary

- Project landing page fully published in [GitHub Pages](#).
- Streaming refactor is partially implemented, progress bar will be simplified once that's complete.
- A blocker is that stopping the RINEX3 conversion seems to make the program hang indefinitely, still looking into the root causes of this. Streaming the data may fix this if it's a memory issue.

Sprint Work Log

Project Landing Page

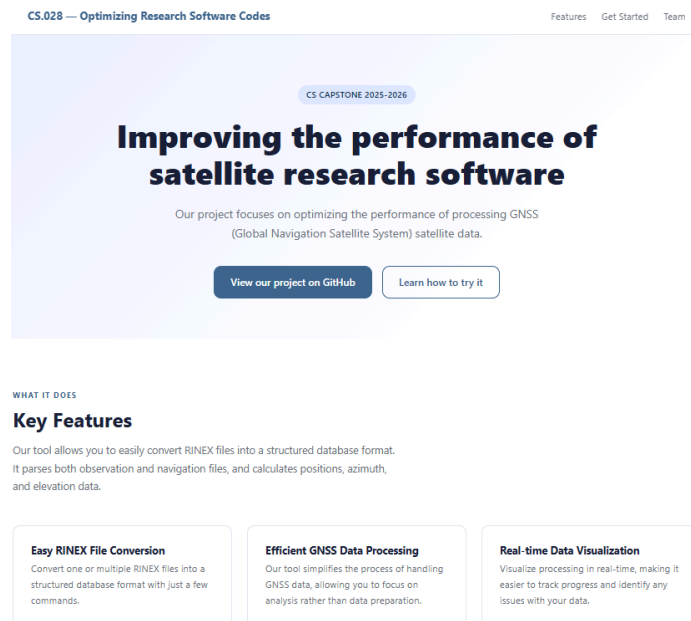
Status: Complete

Author: Kathryn Butler

Reviewers: N/A

Source Link: [GitHub Pages](#)

Supporting Artifact:



Notes: All images have alt text and the page can be navigated with keyboard shortcuts and has ARIA labels for accessibility. I also completed and presented the first draft of our project poster, and will be continuing to work on that as we finish optimizations.

Optimized calculate_positions() Logic

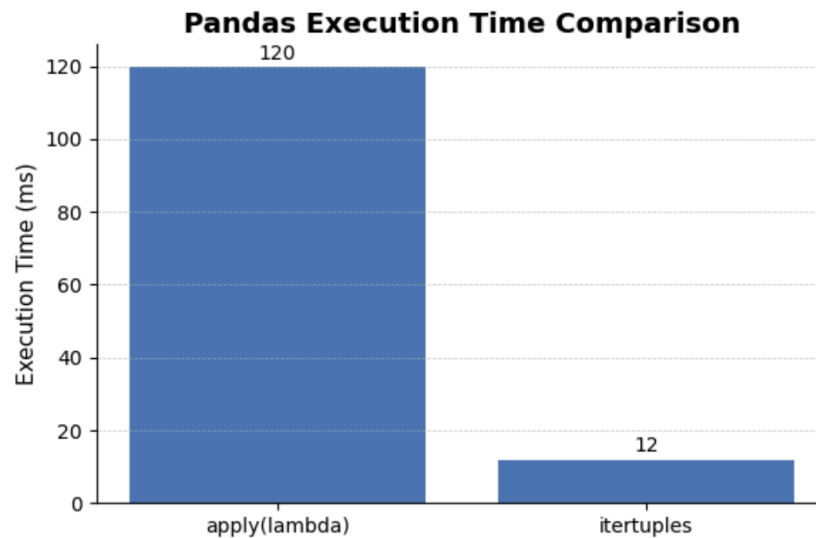
Status: In progress (90% complete)

Author: Joseph Schaab

Reviewers: N/A

Source Link: [Commit 72e8934](#)

Supporting Artifact:



Comparison of Pandas Methods

Notes: Eliminating `apply(lambda)` from `calculate_positions()` greatly reduces the amount of memory the program needs to manage. Inefficient calculation design from the original `calculate_positions()` is significantly responsible for the [rnx2db.py](#) poor runtime. Implementation still requires some internal adjustments to match the original file environment.

Streaming Refactor

Status: In progress (50% complete)

Author: Kathryn Butler

Reviewers: N/A

Source Link: [Commit 9bdb23f](#)

Supporting Artifact: in-progress implementation to replace the full in-memory load with chunked streaming.

Notes: The current progress bars (calculate and write) will be merged into one that tracks the chunks' completion. This should be much easier on memory and timing, especially because the progress bar only has to check in with each chunk instead of with all of the writes at once.

RINEX3 Conversion Hang

Status: In progress (30% complete)

Author: Kathryn Butler

Reviewers: N/A

Source Link: [Commit 9bdb23f](#)

Supporting Artifact: The previous fix stopped the conversion and instead made the program seem to hang indefinitely. This commit reverted the changes until a better way to stop the conversion is found.

Notes: Converting to RINEX2 loses much more data than just keeping RINEX3, so hopefully this fix will allow for the researchers to learn more things with more data.

Risk and Quality Tracking

Bug Count: 11 ([Link to issue query](#))

Top Bugs:

- [Position Variable Mutation](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Identified, waiting on meeting
 - **Actions:** Meet with Dr. Park to ask about the potential calculation issue.
- [Treating Keys as Positions and Using Argsort](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Series.getitem treating keys as positions is deprecated. Look up how to solve it.)
- [Incorrect conversion from float to datetime](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Research into the line causing the problem. Determine the input causing the problem, and correct the datetime variable.)
- [Log.txt hardcoded](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Log.txt seems to be announcing a preset number of cores to calculate. Figure out why it is set to only be 8 cores. Find out if you can manually adjust the core count to be something other than just 8, then add the text to the function dynamically.)

- [Inconsistent timing data](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Create a smaller test data pool, run the tests multiple times to determine if the size of the file increases the likelihood of failed parsing attempts.
- [Xarray.merge conflict](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Set compat explicitly to something other than 'override', then test output for clarity. (Previous goal: Research what "compat" does, and either set it explicitly or use the new combine_kwargs_defaults.)

Next Sprint Plan

- Complete the streaming refactor. Finish the chunked calculations and simplify the progress bar. Source of truth: [Commit 9bdb23f](#)
- Resolve position variable mutation. Meet with Dr. Park to review the code to check if there is an error. Source of truth: [Issue 15](#)
- Stretch goal: Allow users to drop RINEX files into a folder and have the program detect it from there instead of manually updating the config.yaml file. Source of truth: [Issue 16](#)

Team Process Reflection

In this sprint, solo ownership of most of the work allowed for fast decision-making, but it also meant that code review and overall knowledge-sharing between teammates was minimal. With Dr. Park's input of the smaller files, this may make it easier, especially since the runtimes of the program will decrease dramatically which allows for easier testing of improvements.

The RINEX3 conversion fix did not work initially, and led from one failure to another. We will continue to persevere through this and find a way to retain all of the data while optimizing the program.